

Section 9 — Special Data Types

Concept Notes

Pydantic provides built-in support for many real-world data types beyond basic Python types such as `str`, `int`, `float`, and `bool`.

These special types can validate common data formats such as email addresses, URLs, UUIDs, dates, times, IP addresses, and secret values.

Email Validation

`EmailStr` is used to validate email addresses.

It checks whether the input follows a valid email format. This is useful for registration forms, login systems, contact forms, and API request models.

Email validation requires the `email-validator` package.

URL Validation

Pydantic provides URL types such as `HttpUrl`.

It checks whether the input is a properly structured HTTP or HTTPS URL.

This is useful for:

- Website links
- Portfolio URLs
- API endpoints
- Callback URLs

UUID

A UUID is a unique identifier commonly used for users, orders, transactions, and database records.

Pydantic works with Python's `UUID` type and can validate UUID strings automatically.

Date and Time

Pydantic supports Python date and time types such as:

- `date`
- `time`
- `datetime`
- `timedelta`

String values in valid formats can be parsed into Python date and time objects.

IP Address Validation

Pydantic can validate IPv4 and IPv6 addresses using Python's `ipaddress` types.

This is useful for network applications, server configuration, logging systems, and security-related data models.

Secret Values

Sensitive information such as passwords, tokens, and API keys should not be displayed directly in logs or model representations.

Pydantic provides:

- `SecretStr`
- `SecretBytes`

These types hide the actual value when the model is printed. The original secret can still be accessed explicitly when required.

Important Points

- Pydantic supports many specialized data types.
- `EmailStr` validates email addresses.
- `HttpUrl` validates HTTP and HTTPS URLs.
- `UUID` validates unique identifiers.
- `date` stores calendar dates.

- time stores time values.
- datetime stores both date and time.
- timedelta represents a duration.
- IP address types validate IPv4 and IPv6 addresses.
- SecretStr hides sensitive string values.
- SecretBytes hides sensitive byte values.
- Special data types reduce the need for manual validation code.
- Invalid special-type values raise a ValidationError.

Syntax

Email

field_name: EmailStr

URL

field_name: HttpUrl

UUID

field_name: UUID

Date and Time

date_field: date

time_field: time

datetime_field: datetime

duration_field: timedelta

IP Address

field_name: IPvAnyAddress

Secret Value

password: SecretStr

api_key: SecretStr